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SYMBIOSIS

INSTITUTE OF TECHNOLOGY, PUNE

Project-based Learning I

Project Title: Sales and Revenue Analysis for Retail Business

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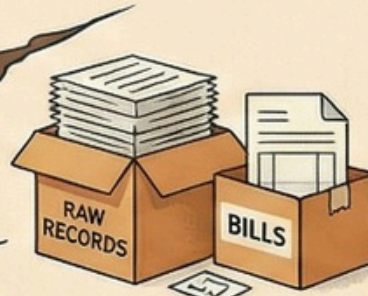
Under the Guidance of
Dr. Anuradha Pillai

SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

2025-26

Visionary Retail: Small Business, Big Data

The Challenge of Manual Management



The Cost of Guesswork
Purchasing based on instinct leads to expired unsold items and essential product stockouts.



Unexplained Low Profits
Without analysis, shopkeepers struggle to identify why monthly earnings do not meet expectations.



The Power BI Solution

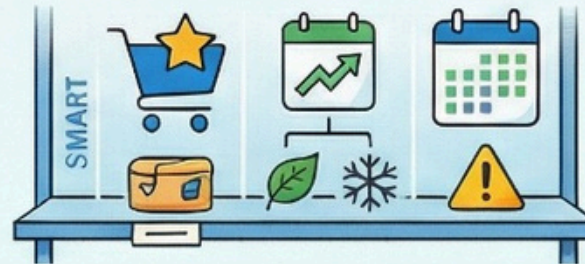
Converting Data to Intelligence

The system transforms simple sales records into clear visualizations of business performance.



Predictive Stocking Insights

Analytics reveal top-selling products, seasonal trends, and items causing financial losses.



Driving Global Goals (SDG 8)

Empowering merchants with data supports Decent Work and Economic Growth.



Traditional Retail vs. Visionary Retail (Power BI)

 Inventory Strategy Guesswork & Instinct	 Data-Backed Stocking
 Product Lifecycle High Waste/Expired Goods	 Optimized Optimized Stock Levels
 Decision Basis Confusion at Month-End	 Smart Business Decisions



- Converts raw retail data into actionable business insights
- Analyzes revenue, profit, and sales trends
- Identifies top-selling and slow-moving products
- Detects seasonal demand patterns for better forecasting
- Enables data-driven inventory and stocking decisions
- Uses Market Basket Analysis to identify frequently bought-together products and support cross-selling
- Integrates ML models (XGBoost, Prophet) for predictive insights
- Supports business growth and improved profitability

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Introduction



Presented by: Aruna Gupta & Anuradha Pillai
Guided by: Dr. Anuradha Pillai
Academic Year: 2025-26

The Problem:



Massive transactional data, but untapped for strategy.

but untapped for strategy.



The Strategic Solution:

Leveraging Power BI: Data to Knowledge

The Outcome



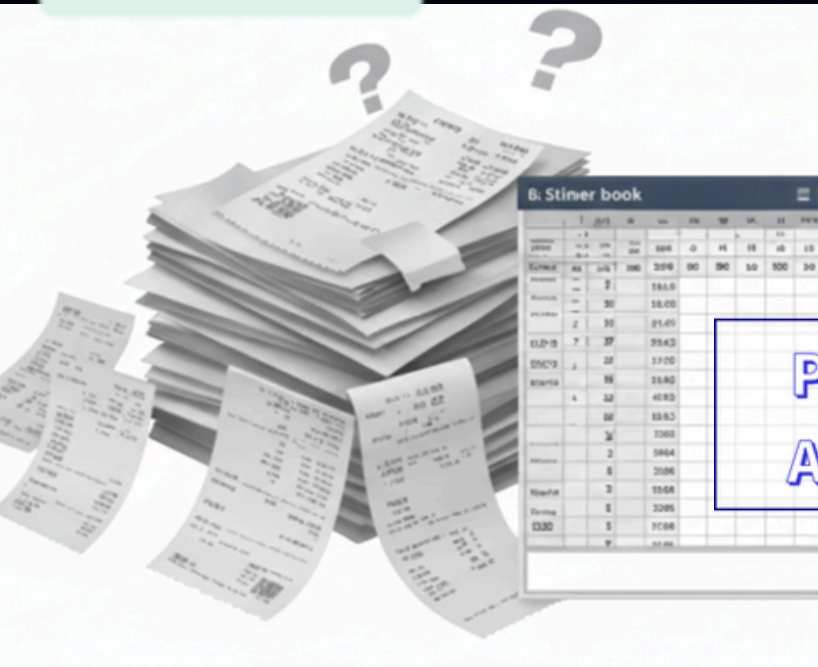
Actionable Insights & Visual Intelligence

Global Impact- SDG 8

Driving Sustainable Economic Growth by transforming raw retail data into actionable business intelligence.

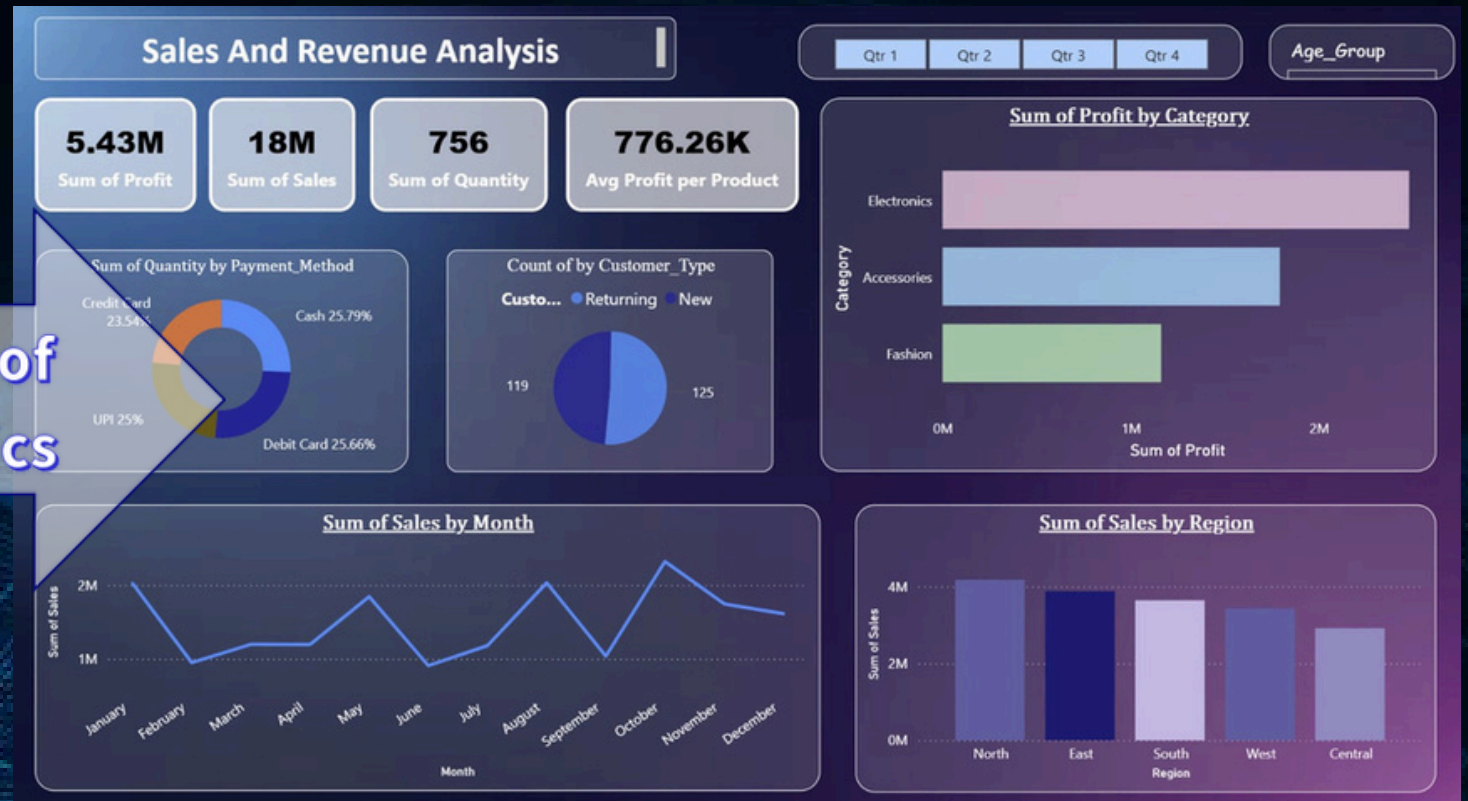
Motivation- Bridging the Visibility Gap

The Problem:



Unorganized data leads to guesswork, stock shortages, and missed revenue.

Power of Analytics



SDG 8 Alignment: Promoting Economic Productivity through Data-Driven Strategies

Statement of Problem/Scope

The Problem:

Data Underutilization: Retailers collect vast amounts of data, rarely using it for strategy.

Lack of Visibility: Struggle to identify trends, top products, and seasonal changes.

Operational Risks: Leads to overstocking, stock shortages, and inefficient planning.



Project Scope:

Data Processing: Collect, clean, and transform raw transactional data.

Visual Intelligence: Develop interactive Power BI dashboards for key metrics.

Predictive Analytics: To predict revenue and demand.

Strategic Support: Deliver actionable insights for informed decisions.

Objectives of Study

Data Refinement

To collect, clean, and preprocess raw sales records to ensure high data quality and accuracy for analysis.

Trend Visualization

To visualize key metrics, such as top-selling products and monthly revenue trends, using interactive charts to reveal hidden purchasing patterns

Predictive Insights

To explore time-based forecasting methods for estimating future revenue and anticipating customer demand fluctuations.

Strategic Decision Support

To provide actionable recommendations for inventory planning and marketing strategies based on data-driven evidence.

SDG Integration

To demonstrate how technological upgrading via Power BI contributes to enhanced economic productivity and sustainable business growth (SDG 8.2).

Existing Systems



Basic Spreadsheets: Tools like Excel used for simple storage, which lack real-time visualization and predictive capabilities for forecasting.



Manual Ledgers: Traditional handwritten records which are prone to human error and offer no analytical value.



Standard POS Systems: Transactional systems that record sales but do not "clean" or "analyze" the data to provide strategic business insights.

Comparison of Existing Systems

Feature	Manual Ledger	Basic POS	Proposed System
Data Cleaning	X	X	✓(Standardized)
Visual Dashboard	X	Static	✓(Interactive)
Future Forecasting	X	X	✓(Time-Based)
SDG 8 Support	X	X	Direct (Economic Growth)

Alignment with SDG 8 – Decent Work & Economic Growth

- ✓ Improves business profitability and economic growth
- ✓ Enables data-driven decision-making for retailers
- ✓ Reduces inventory losses and operational inefficiencies
- ✓ Supports small and local businesses in scaling operations
- ✓ Enhances productivity through predictive analytics
- ✓ Promotes efficient use of resources and reduced waste
- ✓ Contributes to sustainable and long-term business growth



Prophet: Sales Forecasting Model

Sales
Prediction

=

Trends

+

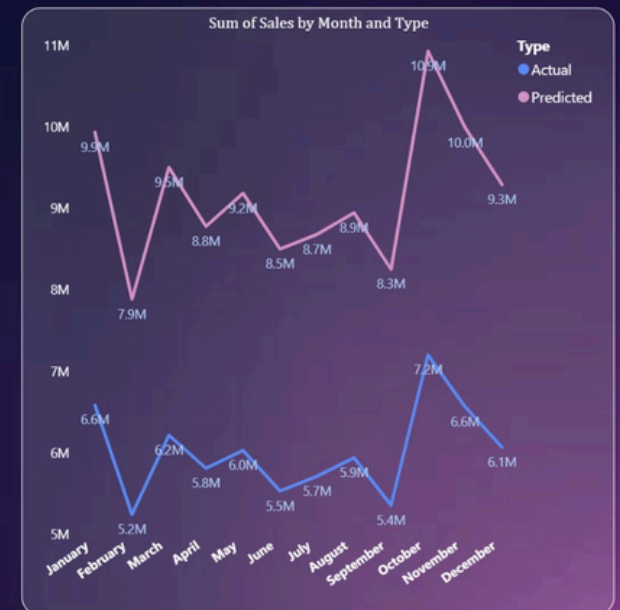
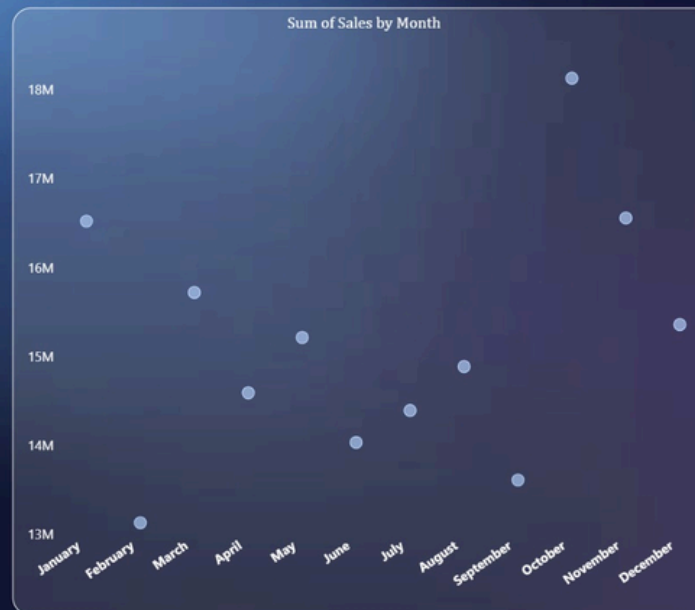
Seasonality

+

Holidays

- Time-series model for sales forecasting
- Developed by Facebook
- Captures trend & seasonality
- Works on date-based data
- Predicts future demand
- Easy to use, minimal tuning

PREDICTED SALES Through Prophet Model



XGBoost: Regression Model for Predictive Analysis

Previous
Errors

+

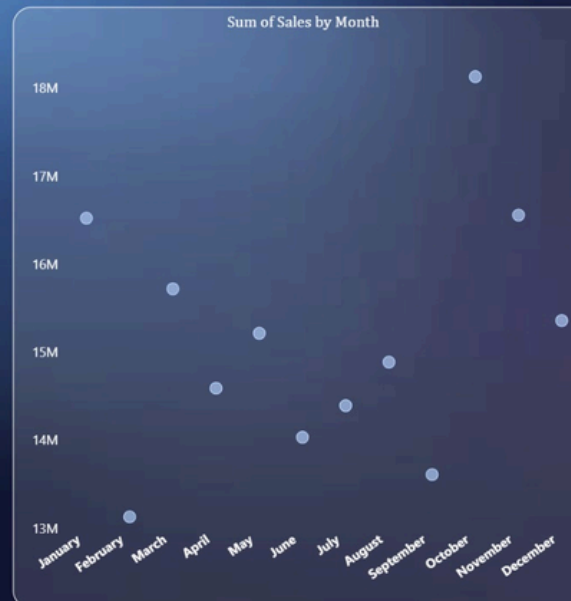
New Trees



Better
Prediction

- Advanced ML algorithm
- Used for prediction tasks
- Handles complex data patterns
- Uses multiple factors (price, region, etc.)
- Provides high accuracy

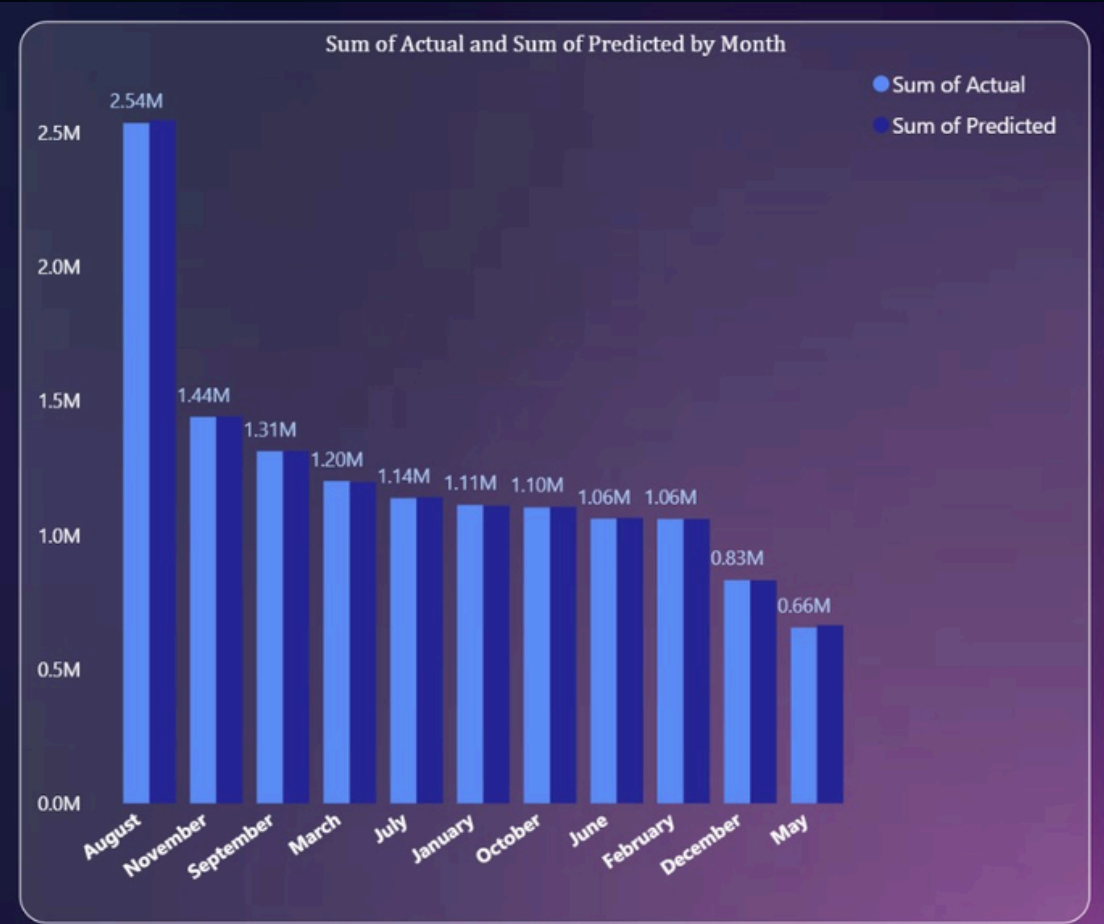
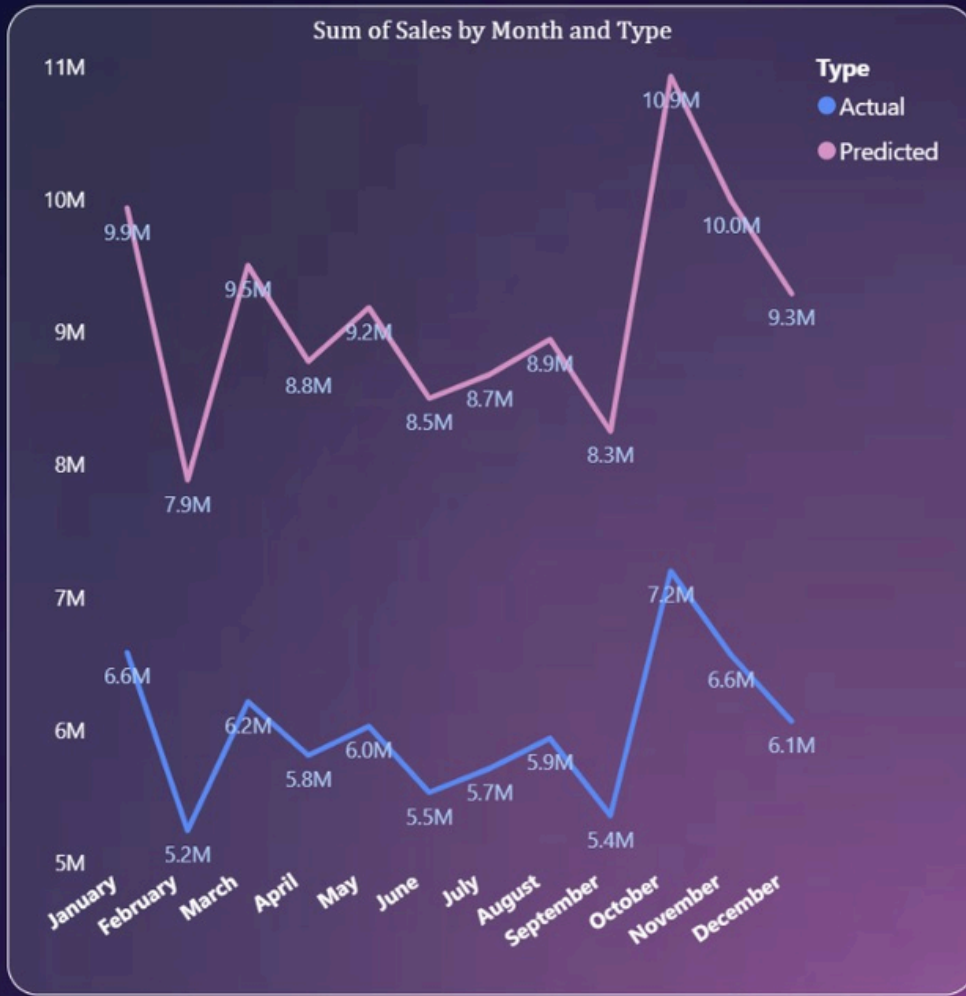
PREDICTED SALES Through XGboost Model



Prophet VS XGBoost

Feature	XGBoost	Prophet
Type	Machine Learning	Time Series Model
Input	Multiple features	Only Date + Sales
Handles Seasonality	✗ (manual)	✓ automatic
Accuracy	Very high	Moderate
Interpretability	Medium	High
Best Use	Complex datasets	Time-based forecasting

Prophet VS XGBoost



Market Basket Analysis

Transactions



Patterns



Product Associations

- Identifies frequently bought-together items
- Uses association rules (Apriori algorithm)
- Helps in cross-selling & product bundling
- Improves store layout & product placement
- Increases average order value
- Enhances customer buying insights

Market Basket Analysis Dashboard

0.82

Max of confidence

1.04

Max of lift

antecedents	consequents	confidence	lift
Bag	Watch	0.74	1.01
Headphones	Watch	0.74	1.01
Laptop	Watch	0.75	1.02
Shoes	Watch	0.74	1.02
Bag	Shoes	0.75	1.02
Watch	Shoes	0.75	1.02
Bag	Mobile	0.75	1.02
Clothes	Mobile	0.75	1.02
Headphones	Mobile	0.75	1.02
Headphones	Laptop	0.82	1.04
Watch	Laptop	0.81	1.02
Bag	Headphones	0.74	1.01
Laptop	Headphones	0.76	1.04
Mobile	Headphones	0.74	1.02

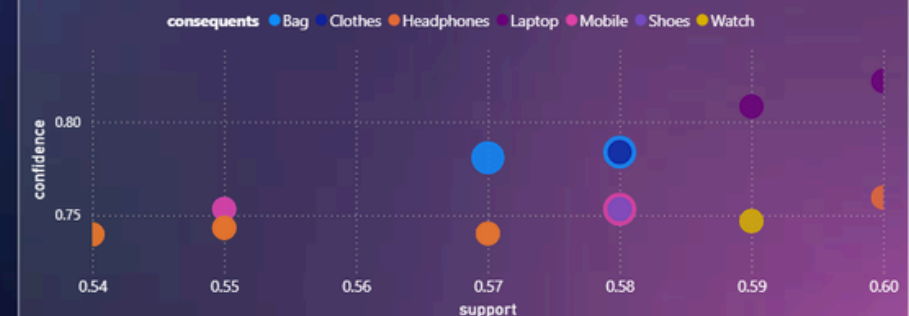
antecedents

Select all

Sum of confidence by consequents



Sum of lift by consequents, support and confidence



Conclusion & Future Impact

Key Achievements

- Converted raw retail data into actionable insights
- Identified sales trends, top products, and demand patterns
- Applied Prophet & XGBoost for accurate forecasting
- Used Market Basket Analysis for better cross-selling strategies
- Enabled data-driven decision-making in inventory and sales



Future Impact

- **Smarter Inventory Management:** Reduce stockouts & overstocking
- **Revenue Growth:** Improve sales through predictive insights & bundling
- **Customer Insights:** Better understanding of buying behavior
- **Scalability:** Can be expanded to larger datasets and real-time analytics
- **Sustainable Growth:** Supports long-term, data-driven retail success (SDG 8)

References

- [1] United Nations, “Transforming our world: The 2030 Agenda for Sustainable Development,” United Nations, 2015. [Online]. Available: <https://sdgs.un.org/2030agenda>
- [2] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Burlington, MA, USA: Morgan Kaufmann, 2011.
- [3] Microsoft, “Power BI documentation,” Microsoft Learn. [Online]. Available: <https://learn.microsoft.com/en-us/power-bi/>
- [4] Deloitte, “Global Powers of Retailing,” Deloitte Insights, 2023. [Online]. Available: <https://www2.deloitte.com/>

The background is a dark blue gradient. It features a faint, glowing hexagonal grid pattern that resembles a honeycomb or molecular structure. Overlaid on this grid are several concentric, wavy lines that create a sense of depth and movement, similar to ripples on water or sound waves. The overall effect is a complex, layered, and futuristic aesthetic.

Thank You